

Deep Learning Specialist | Machine Learning Engineer Aspirant
Machine Learning Engineer aspirant with over 4 years of experience in algorithm development and data analysis; spearheaded research initiatives that enhanced patient-clinician interactions within therapeutic web-based platforms. Achievements include reducing audio latency from 400ms to 80ms for the Crowd Mics app, resulting in improved user experience on major platforms such as Google Play Store and Apple App Store. Holds a Master of Science in Artificial Intelligence from Queen Mary University of London.

SKILLS

Tools and Languages	Python libraries (PyTorch, torchaudio, Tensorflow, Librosa,NumPy, Scikit-Learn, Matplotlib, Tensorboard, imbalanced-learn, networkx), Databases (Mysql, MariaDB), Javascript(React, React-native, Nodejs).
Quantitative Research	Machine Learning, Deep Learning, Neural Networks (CNNs,RNNs, generative models, transformers, autoencoders), Linear Algebra, Information Retrieval, Natural Language Processing(Text classification, Neural Machine Translation, Named entity recognition)
Technologies/Frameworks	Linux, Git, Material UI, Hugging Face, Tailwind CSS, Open AI, Langchain, Large Language Models

TECHNICAL EXPERIENCE

AI Research Engineer Cognitive Science Group <i>Queen Mary University of London (QMUL)</i>	May 2022 - November 2024 <i>London</i>
- Initiated research initiatives centered on RemoteDialogPlus and Multi-Model Disfluency Detection, resulting in the identification of over 3 key areas for machine learning application within therapeutic web-based platforms. - RemoteDialog+ is the first approach specifically implemented for routine patient-clinician meetings that are therapeutically effective using Machine learning. - Collected and manipulated complex medical datasets, identifying three key areas where data mining and machine learning-based solutions were applied and applied in web-based applications. - Designed machine learning algorithms within the RemoteDialog+ project, enhancing therapy session quality for the East London NHS Foundation Trust clinical group by 20% as measured by clinicians. - Developed the RemoteDialog+ platform, now being piloted by the East London NHS Foundation Trust, projected to serve 50+ NHS trusts across the United Kingdom upon full rollout. - Engineered a multi-modal disfluency detection model specifically for the German language by integrating transcripts, audio samples, and hand gestures. - Realized state-of-the-art performance utilizing BERT frameworks and audio transformer architectures tailored for the German dialect, resulting in exceptional metrics including a remarkable accuracy rate of 0.98 and F1 score at 0.41.	
Full-stack Developer <i>Antares.Tech</i>	May 2019 - February 2020 <i>Noida, India</i>
- Delivered tailored application development services focusing on both business-to-business (B2B) and business-to-consumer (B2C) platforms; findings were used to address core customer needs resulting in enhanced product-market fit. - Most of the contribution was on an app called Crowd Mics which is available on Google Play Store and Apple App Store for mobile devices. - Collaborated on innovative solutions targeting audio processing delays; achieved a reduction of latency from 400ms down to just 80ms, dramatically improving real-time interaction for users on both iOS and Android platforms. - Launched a CI/CD pipeline with Jenkins and Docker, achieving 99.99% uptime and reducing deployment time. - Developed and deployed mobile applications for"mobile operating systems utilizing React-Native, achieving a 100% shared UI codebase that enhanced development efficiency across device ecosystems.	

EDUCATION

Master of Science in Artificial intelligence , <i>Queen Mary University of London, London</i>	September 2021 — September 2022
Bachelor of Technology(Computer Science) , <i>Indian Institute of Information Technology, Guwahati India</i>	June 2015 — May 2019

PUBLICATION

Ella Cullen, Patrick Healey, Paraskevi Argyriou and Suyog Pipliwal. Characteristics of Common Experimental Dialogue Tasks: a Systematic Review & Taxonomy